

FINITE DYNAMICS OF THE IDEMPOTENT-LIFTING CUBIC $3x^2 - 2x^3$ MODULO POWERS OF TWO

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ABSTRACT. Burban and Drozd explicitly record $3x^2 - 2x^3$ in a known idempotent-lifting construction with quadratic error improvement; the cubic and that mechanism receive no contribution credit here. We instead determine its complete temporal and one-step inverse atlas on $\mathbb{Z}/2^n\mathbb{Z}$. A parity-selected endpoint error has a valuation that doubles at every epoch, which gives every pointwise entry time, every temporal shell, the sharp height $\lceil \log_2 n \rceil$, and the two recurrent points. The inverse problem retains information discarded by this clock. In output valuation $2v$, its normalized odd unit lies in one prescribed class modulo eight, with different classes for $v = 1$ and $v \geq 2$. An exact branchwise bit lift then gives every target fibre, including the separate quotient boundaries $N = n - 2v = 1, 2$, and yields a closed image-size formula. A deterministic audit checks every state and target through $n = 17$ and the normalized-unit lanes $1 \leq v \leq 6$, $1 \leq N \leq 11$, totaling 2,563,880 exact assertions.

1. PRIOR BOUNDARY AND THE SCOPED-ATLAS THEOREM

Burban and Drozd's Lemma A.4 recalls a known result, with a pointer to classical background, giving integer polynomials that improve an approximate idempotent modulo successive powers of an ideal; it explicitly lists

$$G_1(x) = 3x^2 - 2x^3$$

as an example [1]. We use this as a direct prior record, not as an origination claim. Thus neither the literal cubic, its idempotent-lifting role, nor quadratic error improvement is new here. We ask the narrower finite question: what are all entry-time shells, image classes, and one-step fibres of this fixed map modulo 2^n ?

For $n \geq 1$, write

$$F_n : \mathbb{Z}/2^n\mathbb{Z} \longrightarrow \mathbb{Z}/2^n\mathbb{Z}, \quad F_n(x) = 3x^2 - 2x^3.$$

Select the endpoint error by parity,

$$(1) \quad e(x) = \begin{cases} x, & x \equiv 0 \pmod{2}, \\ 1 - x, & x \equiv 1 \pmod{2}, \end{cases}$$

and use the truncated convention $v_2(0 \bmod 2^n) = n$. Let $\tau_n(x)$ be the first entry time into $\{0, 1\}$.

Theorem 1 (temporal and one-step inverse atlas). *For every $n \geq 1$, the following statements hold.*

(A) *For all $x \in \mathbb{Z}/2^n\mathbb{Z}$ and $t \geq 0$,*

$$(2) \quad v_2(e(F_n^t(x))) = \min\{n, 2^t v_2(e(x))\}.$$

The reached endpoint is $x \bmod 2$, and

$$(3) \quad \tau_n(x) = \min\{t \geq 0 : 2^t v_2(e(x)) \geq n\}.$$

For $t \geq 0$,

$$(4) \quad A_{n,t} := \#\{x : \tau_n(x) \leq t\} = 2^{n - \lceil n/2^t \rceil + 1}.$$

Consequently the exact shell polynomial is

$$(5) \quad D_n(z) = A_{n,0} + \sum_{t=1}^{M_n} (A_{n,t} - A_{n,t-1})z^t, \quad M_n = \lceil \log_2 n \rceil,$$

and 0, 1 are the only recurrent states.

(B) *Let $y \notin \{0, 1\}$ and reflect it into the even basin by*

$$y^{\flat} = \begin{cases} y, & y \equiv 0 \pmod{2}, \\ 1 - y, & y \equiv 1 \pmod{2}. \end{cases}$$

Then $y \in \text{im } F_n$ if and only if $v_2(y^b) = 2v < n$ for an integer $v \geq 1$ and, on putting $N = n - 2v$ and $u = y^b / 2^{2v}$,

$$(6) \quad u \equiv \begin{cases} 1 \pmod{2}, & N = 1, \\ 3 \pmod{4}, & N = 2, \\ 7 \pmod{8}, & N \geq 3 \text{ and } v = 1, \\ 3 \pmod{8}, & N \geq 3 \text{ and } v \geq 2. \end{cases}$$

Every admissible target in this stratum has exactly

$$(7) \quad \#F_n^{-1}(y) = 2^{v+\min(N-1,2)} = 2^{v+\min(n-2v-1,2)}$$

preimages; every inadmissible nonendpoint target has none. The endpoint fibres and total image size are

$$(8) \quad \#F_n^{-1}(0) = \#F_n^{-1}(1) = 2^{\lfloor n/2 \rfloor},$$

$$(9) \quad \#\text{im } F_n = 2 + 2 \sum_{1 \leq v < n/2} 2^{\max(0, n-2v-3)}.$$

The two axes in Theorem 1 are not interchangeable. Part (A) keeps only the error valuation. Part (B) needs the normalized odd unit and even distinguishes the first positive valuation stratum from all later ones. Table 1 records the claim boundary.

TABLE 1. Prior-art subtraction. Every item in the middle column receives zero contribution credit.

Source	Zero-credit prior input	Residual treated here
Burban–Drozd direct record [1]	cubic, idempotent lifting, and quadratic improvement	finite temporal census plus normalized-unit image/fibre atlas
Elementary arithmetic	error factorizations, odd squares modulo eight, divisibility counts	displayed temporal/inverse conjunction, not the ingredients
Generic finite dynamics	orbit, image, and fibre terminology	exact formulas only for this map on $\mathbb{Z}/2^n\mathbb{Z}$

2. EXACT ERROR SQUARING AND THE TEMPORAL CENSUS

The forward law follows from two identities, but we print the count so that the clock convention and the $n = 1$ boundary are explicit.

Proof of Theorem 1(A). Direct expansion gives

$$(10) \quad F_n(x) = x^2(3 - 2x), \quad 1 - F_n(x) = (1 - x)^2(1 + 2x), \quad F_n(1 - x) = 1 - F_n(x).$$

If x is even, then $3 - 2x$ is odd. If x is odd, then $1 - x$ is even and $1 + 2x$ is odd. Parity is preserved, and hence

$$v_2(e(F_n(x))) = \min\{n, 2v_2(e(x))\}.$$

Induction proves (2); the first value n is reached at the time in (3).

Put $a_t = \lceil n/2^t \rceil$. Entry by time t is equivalent to $2^{a_t} \mid e(x)$. The even basin contains 2^{n-a_t} such residues, and reflection $x \mapsto 1 - x$ gives the same count in the disjoint odd basin. This proves (4). Consecutive differences give (5). Every nonendpoint selected error has valuation at least one, so $2^t \geq n$ suffices for absorption; errors of valuation one show that the sharp maximum is M_n . All states reach 0 or 1, and both are fixed, proving the recurrence assertion. When $n = 1$, every state is already an endpoint, $M_1 = 0$, and the same formulas give $A_{1,0} = 2$. $\square$

This section is an exact finite specialization of the zero-credit quadratic improvement mechanism; it supplies context for the inverse atlas rather than a claim to a new lifting iteration.

3. THE NORMALIZED-UNIT LIFTING LAW

For a nonzero even source write $x = 2^v w$, where $v \geq 1$ and w is odd. Then

$$(11) \quad F_n(x) = 2^{2v} h_v(w), \quad h_v(w) = w^2(3 - 2^{v+1}w).$$

The odd factor $h_v(w)$ retains the information that the temporal valuation forgets.

Lemma 2 (branchwise bit lifting). *Fix $v \geq 1$ and let h_v act on the odd residues modulo 2^N . For $N = 1$, its image is the unique odd class and that target has one preimage. For $N = 2$, its image is 3 (mod 4) and that target has two preimages. For $N \geq 3$, the image is 7 (mod 8) if $v = 1$ and 3 (mod 8) if $v \geq 2$; every target in the displayed class has four preimages.*

Proof. For odd w , $w^2 \equiv 1 \pmod{8}$. If $v = 1$, the second factor in (11) is $3 - 4w \equiv 7 \pmod{8}$; if $v \geq 2$, the cubic term is divisible by eight and $h_v(w) \equiv 3 \pmod{8}$. Reduction gives the claimed images and direct fibre counts for $N = 1, 2$.

Suppose $N \geq 3$. Split the odd inputs as $w = r + 4z$ with $r \in \{1, 3\}$ and define the integer-valued function

$$(12) \quad \Phi_{v,r}(z) = \frac{h_v(r + 4z) - h_v(r)}{8}.$$

We prove that $\Phi_{v,r}$ is a permutation modulo 2^k for every $k \geq 0$. For odd w ,

$$\begin{aligned} h'_v(w) &= 6w(1 - 2^v w), \\ h''_v(w) &= 6 - 6 \cdot 2^{v+1} w, \\ h'''_v(w) &= -6 \cdot 2^{v+1}. \end{aligned}$$

Thus $v_2(h'_v(w)) = 1$, $v_2(h''_v(w)) = 1$, and $v_2(h'''_v(w)) \geq 3$. Set $\delta = 4 \cdot 2^j$. The exact cubic Taylor identity, divided by eight, is

$$(13) \quad \frac{h_v(w + \delta) - h_v(w)}{8} = \frac{\delta h'_v(w)}{8} + \frac{\delta^2 h''_v(w)}{16} + \frac{\delta^3 h'''_v(w)}{48}.$$

The first term is 2^j times an odd integer. The last two terms have valuations at least $2j + 1$ and $3j + 5$, respectively. Therefore

$$(14) \quad \Phi_{v,r}(z + 2^j) - \Phi_{v,r}(z) \equiv 2^j \pmod{2^{j+1}}.$$

Starting with the sole residue modulo one, (14) says that the two lifts of every input class hit the two lifts of its output class. Induction proves the permutation assertion.

Take $k = N - 3$. Within either r -branch, z ranges modulo 2^{N-2} , whereas the output beyond its fixed three low bits depends on $\Phi_{v,r}(z)$ modulo 2^{N-3} . Each branch therefore covers the required output class twice; the two branches together give four preimages. This proves the lemma. $\square$

The separate statements for $N = 1$ and $N = 2$ are essential: the four-to-one law begins only at $N = 3$.

4. EVERY-TARGET FIBRES AND IMAGE SIZE

Proof of Theorem 1(B). Equation (11) shows that a nonzero even output has valuation exactly $2v < n$. Conversely, if $2v \geq n$, its output is zero. For $2v < n$, put $N = n - 2v$. Lemma 2 gives exactly the normalized unit conditions in (6).

The source unit w is specified modulo 2^{n-v} , but $h_v(w)$ modulo 2^N depends only on w modulo 2^N . Reduction therefore contributes

$$2^{(n-v)-N} = 2^v$$

high lifts. Lemma 2 contributes $2^{\min(N-1, 2)}$ reduced solutions. Their product proves (7), including $N = 1, 2$.

The zero fibre consists precisely of residues divisible by $2^{\lceil n/2 \rceil}$, and so has $2^{\lfloor n/2 \rfloor}$ elements. The reflection in (10) transfers this entire even-side description to the odd basin and gives the same fibre over one.

Finally, a fixed class modulo eight contains 2^{N-3} odd units when $N \geq 3$, while the $N = 1, 2$ image has one unit. Thus a nonzero stratum in one parity basin has $2^{\max(0, N-3)}$ targets. Summing over $1 \leq v < n/2$, doubling by reflection, and adding the endpoints proves (9). $\square$

For orientation, the fibre spectrum is nonuniform even when all targets are viewed at the same modulus. At $n = 17$, for example, the exact image contains 10,926 targets and the positive fibre sizes are 8, 16, 32, 64, 128, 256, 512. This numerical row illustrates the theorem; it is not used in its proof.

5. EXACT CONTROLS AND SCOPE

The paper-local verifier uses only Python integers. It exhausts every state, orbit, target fibre, and image through $n = 17$. Independently, it exhausts the normalized odd-unit maps for $1 \leq v \leq 6$ and $1 \leq N \leq 11$, including the one- and two-bit boundaries. The frozen transcript contains 2,563,880 exact assertions and has SHA-256

`f5f1884f809110ca8ec3a954af1783c774896708495d626f694bbfb23f7876f1`.

Enumeration supplies counterexample pressure only; Sections 2–4 prove the all- n statements.

The result is confined to the single map $3x^2 - 2x^3$ on $\mathbb{Z}/2^n\mathbb{Z}$. It asserts neither an odd-prime analogue nor a general finite-ring theorem. The bounded source screen did not locate the normalized-unit one-step inverse atlas, but that non-hit is not a novelty, priority, ownership-completeness, or release certificate. An earlier source for the residual formulas would require further subtraction.

DECLARATIONS

Data availability. No external data were used. The paper-local verifier and frozen transcript contain the deterministic exact controls.

Ethics. This mathematical study involved no human participants, animals, personal data, or field intervention.

Author contributions. The anonymous author performed the derivations, exact checks, source-boundary audit, and manuscript preparation.

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External status. This artifact remains `HOLD_EXTERNAL`; it is not cleared for posting, submission, circulation, or author contact.

REFERENCES

- [1] Igor Burban and Yuriy Drozd. Derived categories of nodal algebras. *Journal of Algebra*, 272 (1):46–94, February 2004. ISSN 0021-8693. doi: 10.1016/j.jalgebra.2003.07.025. URL <https://arxiv.org/abs/math/0307060>.